PROJECT REVIEW PRESENTATION

SMART RADAR EQUIPMENT HEALTH MONITORING

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INTRODUCTION

Smart Radar Equipment Health Monitoring is a system designed to monitor the health of electronic equipment. It uses a DHT11 sensor to measure temperature and humidity and an MPU6050 sensor to detect vibration and movement. The ESP32 microcontroller processes the sensor data and identifies abnormal conditions. Green, yellow, and red LEDs indicate the equipment status, while an LCD displays the readings. This system helps detect possible faults early and improves equipment monitoring

PROBLEM STATEMENT

Radar equipment may experience abnormal vibration and changes in temperature during operation. Manual monitoring can make it difficult to identify problems at an early stage. Without continuous monitoring,equipment faults may go unnoticed.

The Main problems are:

i. Difficulty in continuous equipment monitoring.
ii. Delay in identifying abnormal vibration.
iii. Changes in temperature and humidity.
iv. Lack of immediate visual alerts.
v. Risk of reduced equipment performance.

OBJECTIVES

To develop an intelligent monitoring system that detects abnormal vibration and environmental changes in radar equipment, processes sensor data in real time, and provides clear status indications. Monitor.

- Continuous Monitoring Early Fault Detection
- Real-Time Status Indication
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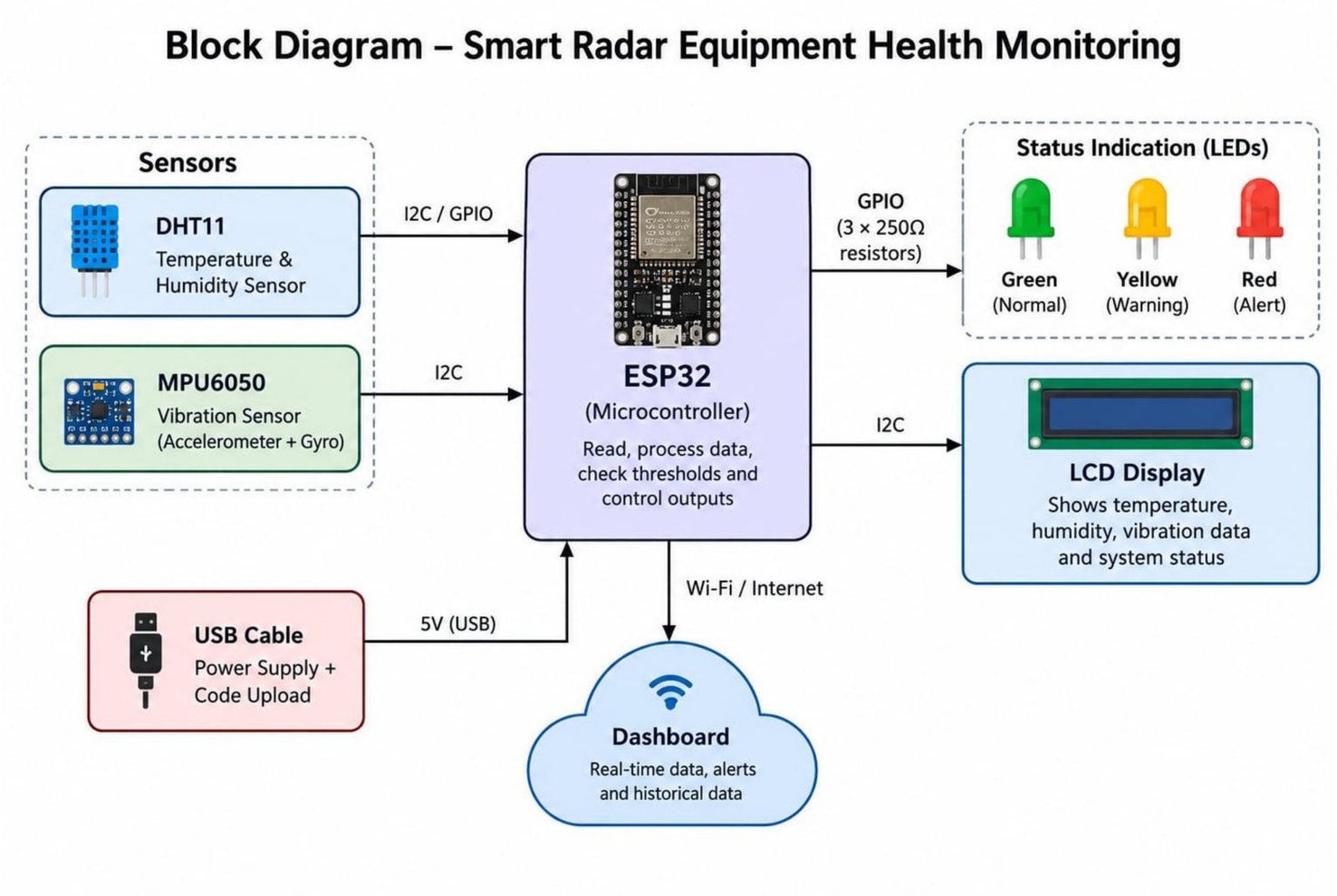
MATERIALS REQUIRED

Component	Purpose
ESP32	Reads and processes sensor data
DHT11	Measures temperature and humidity
MPU6050	Measures acceleration and angular velocity
Green LED	Indicates normal condition
Yellow LED	Indicates warning condition
Red LED	Indicates alert condition
Resistors	Limit LED current
Breadboard and wires	Circuit connections
USB cable	Programming and power

BLOCK DIAGRAM



Sensor Input → Data Processing → Status Output

WORKING PRINCIPLE

1. **Sensor Input:** DHT11 measures temperature and humidity, while MPU6050 measures vibration-related movement.
2. **Data Processing**: ESP32 collects and processes the sensor readings.
3. **Threshold Checking:** The system compares readings with predefined limits to identify abnormal conditions.
4. **Status Indication:** Green, yellow, and red LEDs indicate the equipment status.
5. **Display Output:** The corresponding LED is activated.The LCD displays sensor readings and the current equipment status.

FUTURE SCOPE

- Predictive Maintenance
- IoT-Based Remote Monitoring
- Advanced Vibration Analysis
- Real-Time Fault Detection
- Smart Equipment Monitoring

Proposed Enhancements for the Next Development Stage

CONCLUSION

The proposed project develops a smart system for monitoring radar equipment health. It uses sensors to monitor temperature, humidity, and vibration-related data. The ESP32 processes the readings and indicates equipment status through LEDs and an LCD. The system aims to detect abnormal conditions early and improve equipment monitoring.

- Detect abnormal vibration and temperature changes.
- Improve equipment monitoring through real-time sensor feedback.

Thank You